



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING



LAB MANUAL

CSA1515

CLOUD COMPUTING AND BIG DATA ANALYTICS



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**



CSE DEPARTMENT VISION-MISSION

Vision of the Department

To establish an environment to provide quality education and Inculcate research attributes among computer science engineering graduates through problem solving skills and technological innovations.

Mission of the Department

M1

To create and sustain an academic environment to the highest level in teaching and research by enhancing the knowledge of the faculties and students in technological advancements to solve real time problems.

M2

Providing a suitable environment for the students to develop professionalism with knowledge in Computer Science & Engineering to meet the contemporary industry needs and satisfy global standards.

M3

To facilitate the development of professional behaviour and stronger ethical values so as to work with commitment for the progress of the nation and face challenges with ethical and social responsibility.

Course Outcomes (CO)

CO7	Able to show resource provisioning in the cloud services using VMware workstation and CSP	PO5
CO8	Ability to use the Hadoop Distributed File Systems using Hadoop.	PO5
CO9	Ability to identify suitable Cloud deployment services and BigData analytics to find solutions to real time problems.	PO10
CO10	Ability to identify the solution for real time applications using Cloud and Big Data Analytics	PO10

INSTRUCTIONS FOR THE EXPERIMENTS

1. Students are advised to come to the laboratory on time; those who come after 5 minutes will not be

allowed into the lab.

2. Plan your task properly much before to the commencement, come prepared to the lab with the

synopsis / program / experiment details. Student should enter into the laboratory with:

- Laboratory observation notes with all the details (Problem statement, Aim, Implementation, Steps, Program, Expected Output, etc.,) filled in for the lab session.
- Laboratory Record updated up to the last session experiments and other utensils (if any) needed in the lab.
- Proper Dress code and Identity card.
- Sign in the laboratory login register, write the TIME-IN, and occupy the computer system allotted to you by the faculty.
- Execute your task in the laboratory, and record the results / output in the lab observation note book, and get certified by the concerned faculty.
- All the students should be polite and cooperative with the laboratory staff, must maintain the discipline and decency in the laboratory.
- Computer labs are established with sophisticated and high-end branded systems, which should be utilized properly.
- Students / Faculty must keep their mobile phones in SWITCHED OFF mode during the lab sessions. Misuse of the equipment, misbehaviors with the staff and systems etc., will attract severe punishment.
- Students must take the permission of the faculty in case of any urgency to go out; if anybody found loitering outside the lab / class without permission during working hours will be treated seriously and punished appropriately.
- Students should LOG OFF/ SHUT DOWN the computer system before he/she leaves the lab after completing the task (experiment) in all aspects. He/she must ensure the system / seat is kept properly.



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DO'S AND DON'TS

- ✓ Be on time to lab and maintain silence.
- ✓ Inform the instructor /TA in case of any working environment problem.
- ✓ Be aware of all the safety devices. Even through the instructor and TA will take care of emergencies.
- ✓ Bring all the necessary things required to do laboratory experiments like observation notebook, record notebook and any others.
- ✓ Enter the system number, in-time, out-time register number, name and signature in the students log register.
- ✓ Arrange the seating chairs and system accessories in place at the end of the session.
- ✓ Shut down the system properly and switch off the power switch at the end of the session.
- ✓ Keep your bags in front of the labs empty space.
- ✓ Do not eat, drink, chew gum, smoke or apply cosmetics in the lab.
- ✓ Do not unplug/plug any wires of the system connectivity.
- ✓ Do not use or charge the mobile phone or any electronic gadgets inside the lab.
- ✓ Not to troubleshoot by yourself without knowledge of instructor/TA.
- ✓ Do not open any unnecessary applications in system.
- ✓ Mobiles phones are prohibited inside the lab.
- ✓ Do not share a system to do experiments.

Students strictly follow all the above instructions



SAVEETHA SCHOOL OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**CSA15 – CLOUD COMPUTING AND BIG DATA
ANALYTICS LABORATORY**

LIST OF EXPERIMENTS

SNO	EXPERIMENT NAME	Blooms Taxonomy	CO	PO
1.	Create a simple cloud software application and provide it as a service using any Cloud Service Provider to demonstrate Software as a Service (SaaS).	K03, K04	CO7	PO5
2.	Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS	K03, K04	CO7	PO5
3.	Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS	K03, K04	CO7	PO5
4.	Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS.	K03, K04	CO7	PO5
5.	Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS	K03, K04	CO7	PO5
6.	Create a simple cloud software application for product selling using any cloud service provider to demonstrate saas.	K03, K04	CO7	PO5
7.	Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux).	K03, K04	CO7	PO5

8.	Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.	K03, K04	CO7	PO5
9.	Create a Virtual Hard Disk and allocate the storage using VM ware Workstation	K03, K04	CO7	PO5
10.	Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM	K03, K04	CO7	PO5
11.	Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM	K03, K04	CO7	PO5
12.	Change Hardware compatibility of a VM (Either by clone/create new one) which is already created and configured.	K03, K04	CO7	PO5
13.	Demonstrate Infrastructure as a Service (IaaS) by creating a resources group using a Public Cloud Service Provider (Azure), configure with minimum CPU, RAM, and Storage.	K03, K04	CO7	PO5
14.	Demonstrate Infrastructure as a Service (IaaS) by creating a Virtual Machine using a Public Cloud Service Provider (Azure), configure with required memory and CPU.	K03, K04	CO7	PO5
15.	Demonstrate Infrastructure as a Service (IaaS) by establishing the remote connection, launch the created VM image and run in your desktop.	K03, K04	CO7	PO5
16.	Demonstrate Platform as a Service (PaaS) create and configure a new VM Image in any Public Cloud Service Provider	K03, K04	CO7	PO5
17.	Create a Simple Web Application using Java or Python and host it in any Public Cloud Service Provider (Azure/GCP/AWS) to demonstrate Platform as a Service (PaaS).	K03, K04	CO7	PO5
18.	Demonstrate Storage as a Service (SaaS) create and configure a new VM Image in any Public Cloud Service Provider	K03, K04	CO7	PO5
19.	Create a Storage service using any Public Cloud Service Provider (Azure/GCP/AWS) and check the public accessibility of the stored file to demonstrate Storage as a Service.	K03, K04	CO7	PO5
20.	Database as a Service (DaaS) create and configure a new VM Image in any Public Cloud Service Provider	K03, K04	CO7	PO5
21.	Create a SQL storage service and perform a basic query using any Public Cloud Service Provider (Azure/GCP/AWS) to demonstrate Database as a Service (DaaS)	K03, K04	CO8	PO5

22.	Perform the basic configuration setup for installing HADOOP 2.x like creating the HDUSER and SSH localhost	K03, K04	CO8	PO5
23.	Install Hadoop 2.x and configure the Name Node and Data Node.	K03, K04	CO8	PO5
24.	Launch the Hadoop 2.x and test the Map-Reduce Platform with Hadoop.	K03, K04	CO8	PO5
25.	Launch the Hadoop 2.x and perform Map-Reduce program For a word count problem	K03, K04	CO8	PO5

EXP NO 1: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION AND PROVIDE IT AS A SERVICE USING ANY CLOUD SERVICE PROVIDER TODEMONSTRATE SOFTWARE AS A SERVICE (SAAS).

DATE:

AIM:

To create a simple cloud software application and provide it as a service using any cloud service provider todemonstrate software as a service (saas).

PROCEDURE:

STEP1: GOTO ZOHOCOM

STEP 2: LOGIN TO THE ZOHOCOM

STEP 3: SELECT ONE APPLICATION

STEP 4: ENTER APPLICATION NAME

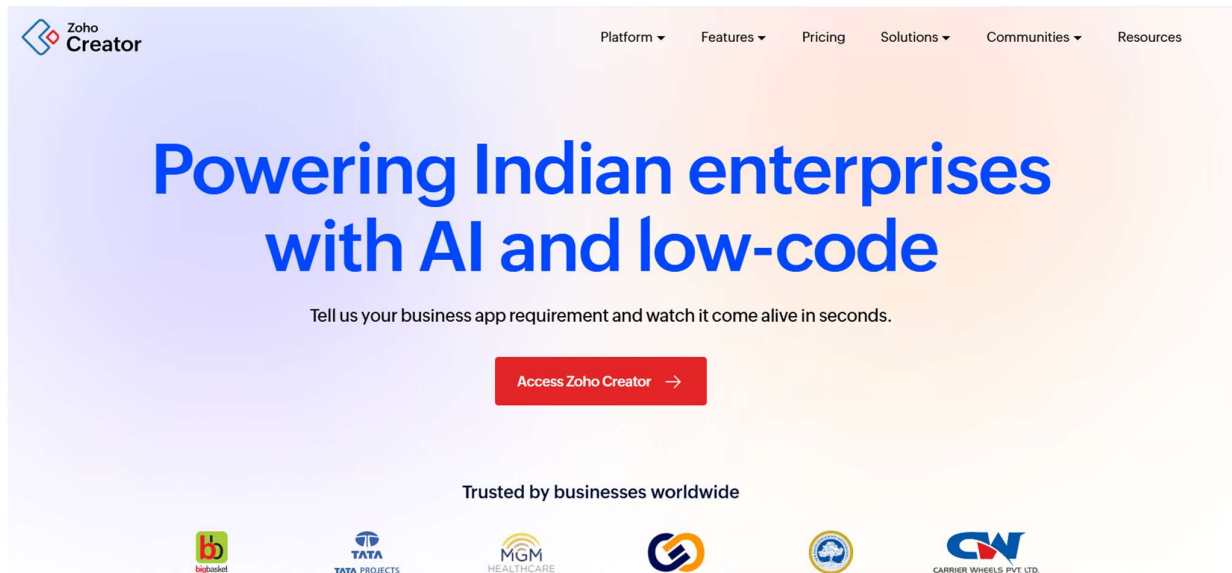
STEP 5: CREATED NEW APPLICATION

STEP 6: SELECT ONE FORM

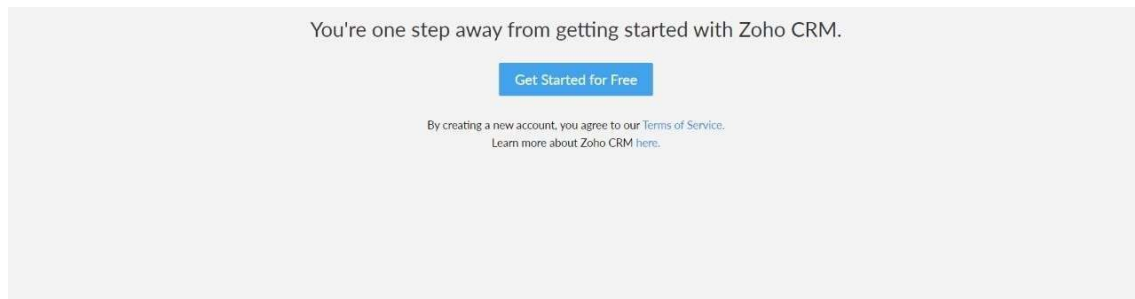
STEP 7: THE SOFTWARE HASE BEEN CREATED.

IMPLEMENTATION:

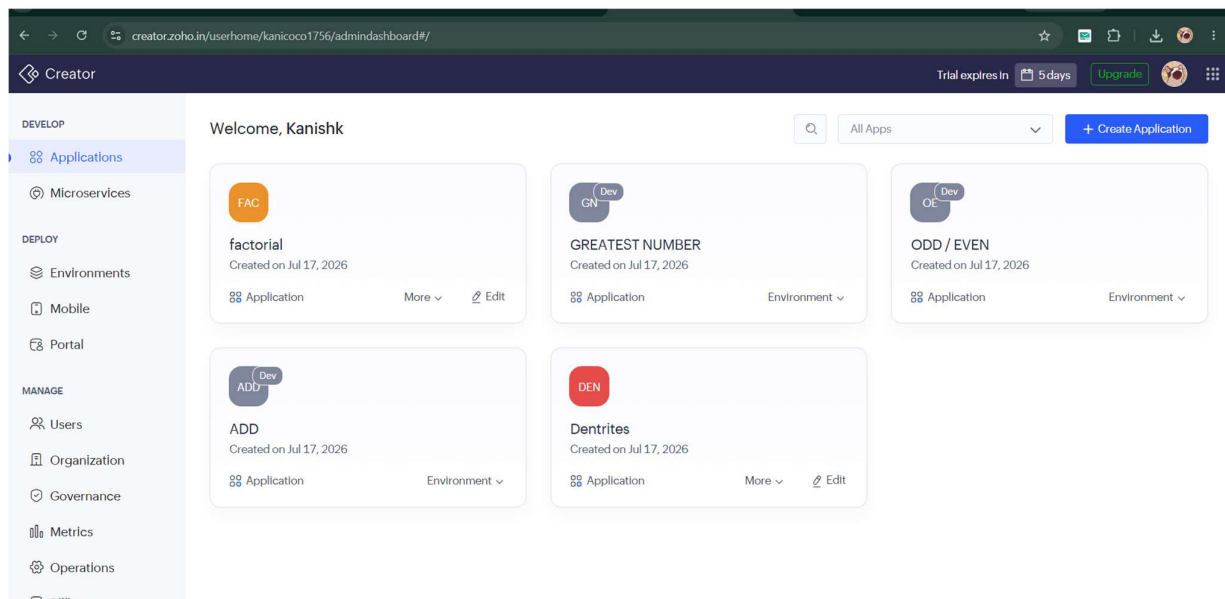
STEP1: GOTO ZOHO.COM



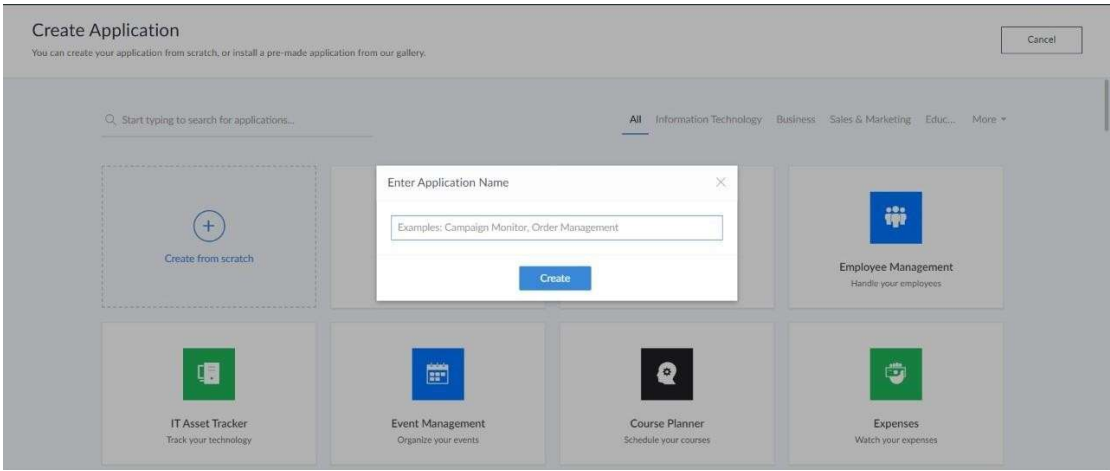
STEP 2: LOGINTOTHE ZOHO.COM



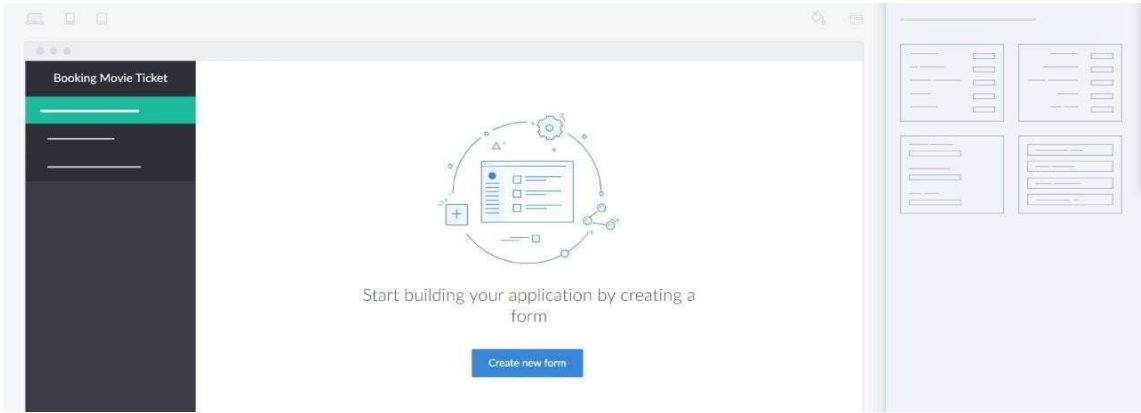
STEP 3: SELECT ONE APPLICATION



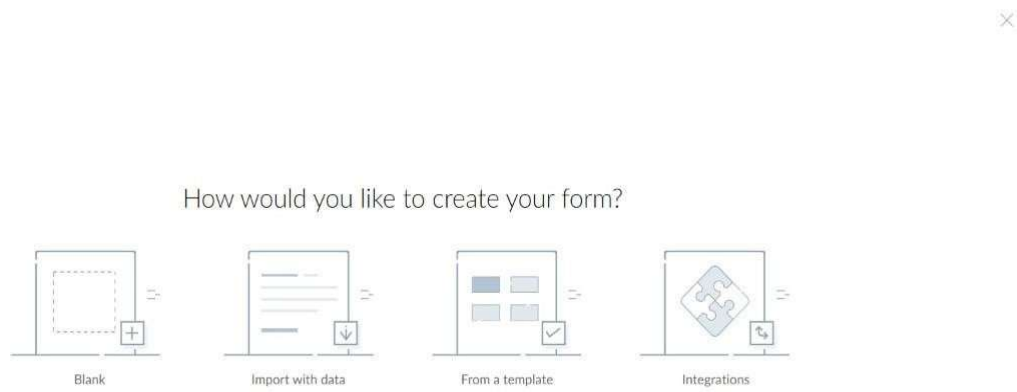
STEP 4: ENTER APPLICATION NAME



STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

123 Number 1

123 Number 2

123 SUM

76

STEP 7: THE SOFTWARE HASE BEEN CREATED.

EXP NO 2: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR FLIGHT RESERVATION SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

DATE:

AIM:

To create a simple cloud software application for flight reservation system using any cloud service provider to demonstrate saas.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as flight reservation system.

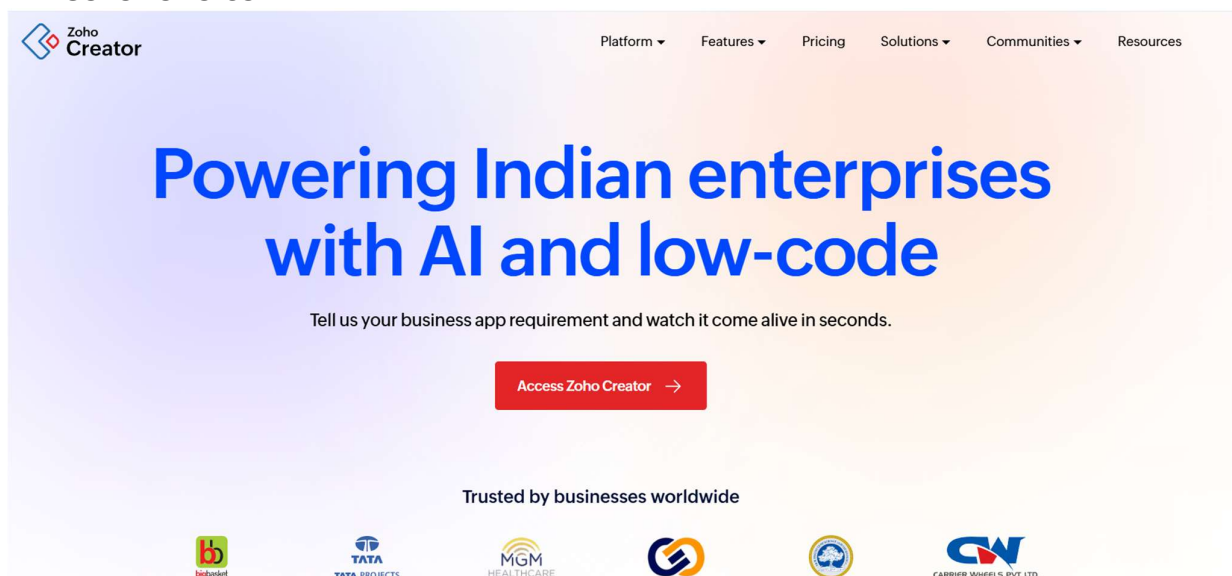
step 5: Created new application flight reservation system.

step 6: Select one form

step 7: The software has been created.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM



STEP 2: LOGINTOTHE ZOHOCOM

You're one step away from getting started with Zoho CRM.

Get Started for Free

By creating a new account, you agree to our [Terms of Service](#).
Learn more about Zoho CRM [here](#).

STEP 3: SELECT ONE APPLICATION

Create Application

You can create your application from scratch, or install a pre-made application from our gallery.

Cancel

Start typing to search for applications...

AllInformation TechnologyBusinessSales & MarketingEduc...More

Create from scratch

Sales Management

More Info | Install this Application

Order Management

Follow your orders

Employee Management

Handle your employees

IT Asset Tracker

Track your technology

Event Management

Organize your events

Course Planner

Schedule your courses

Expenses

Watch your expenses

STEP 4: ENTER APPLICATION NAME

Create from scratch

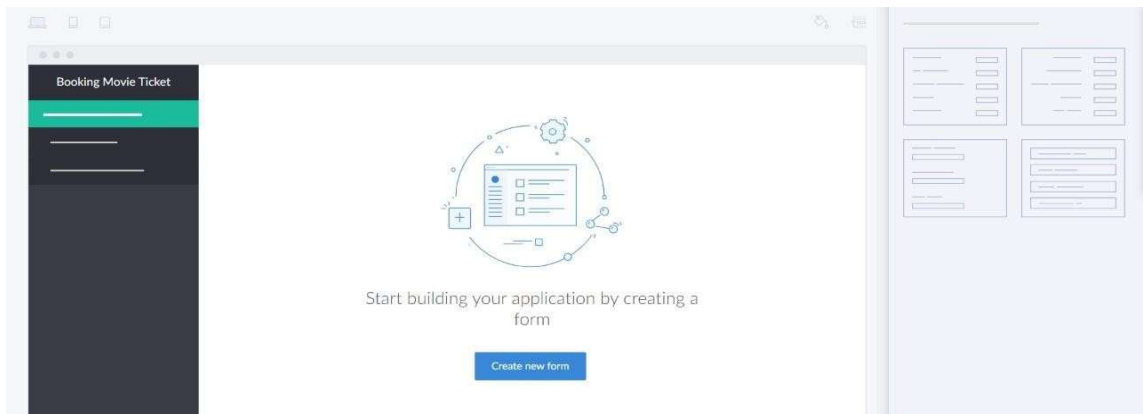
Application Name *

FLIGHT BOOKING SYSTEM

☐ Enable Environment [Learn More](#)

Create Application

STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.

FBS

Pick Your Seat

Name *

First NameLast Name

Mobile Number *

+91

81234 56789

Email *

Date *

dd-MMM-yyyy

Seats *

☐ 1

☐ 2

☐ 3

☐ 9

☐ 17

FLIGHT BOOKING SYSTEM
Pick Your Seat

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Time

Drop Down

Name *

Mobile Number *

Email *

Date *

Your notes content will be shown here

Seats *

Your notes content will be shown here

29

Field Properties

Field name

Name

Field link name

Name

View Field References

Validation

Mandatory

Display Fields

Prefix

First Name

Last Name

Suffix

Booked Seats

	Name	Seats	Email	Mobile Number
29-Jul-2026	KANISHK C	17 5 12	kanicoco1756@gmail.com	+911234567890

EXP NO 3: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR PROPERTY BUYING & RENTAL PROCESS (IN CHENNAI CITY) USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

DATE:

AIM:

To Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as property buying & rental.

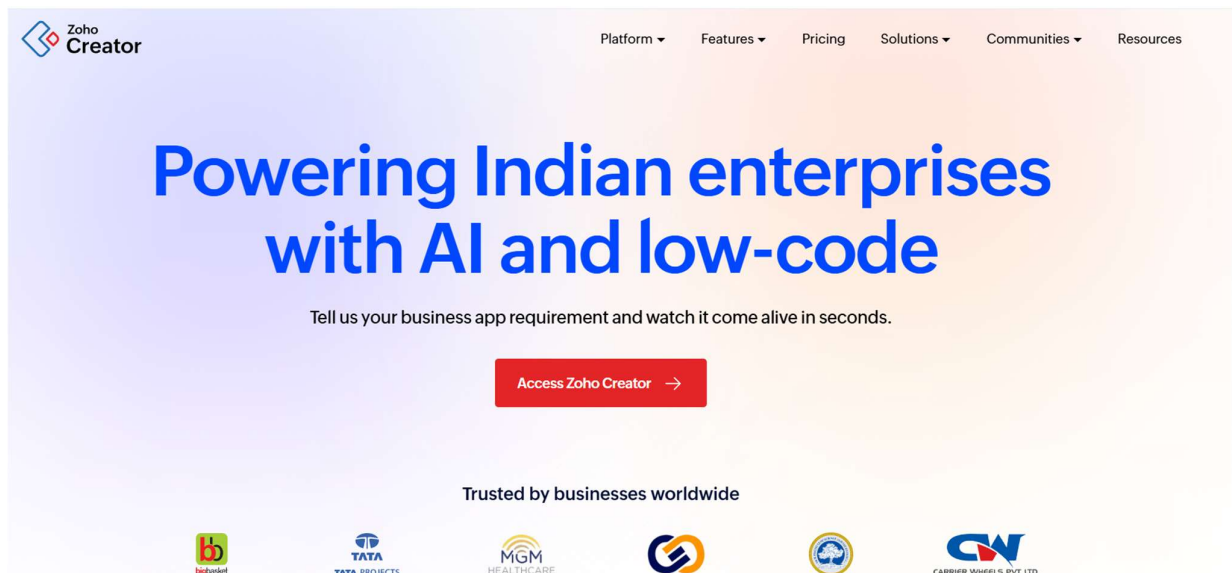
step 5: Created new application as property buying & rental.

step 6: Select one form

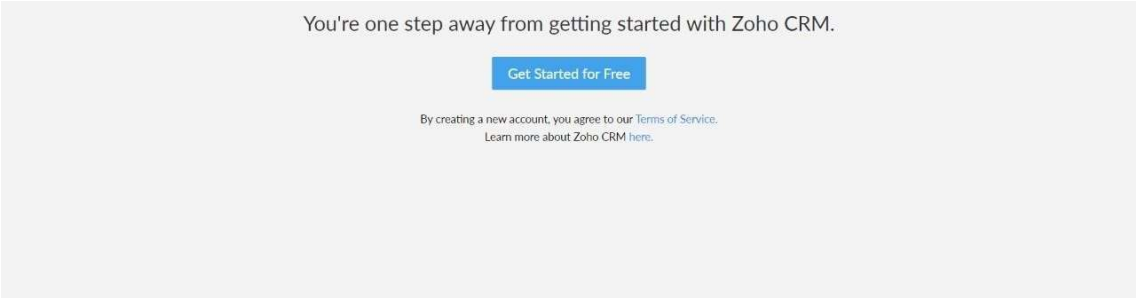
step 7: The software has been created.

IMPLEMENTATION:

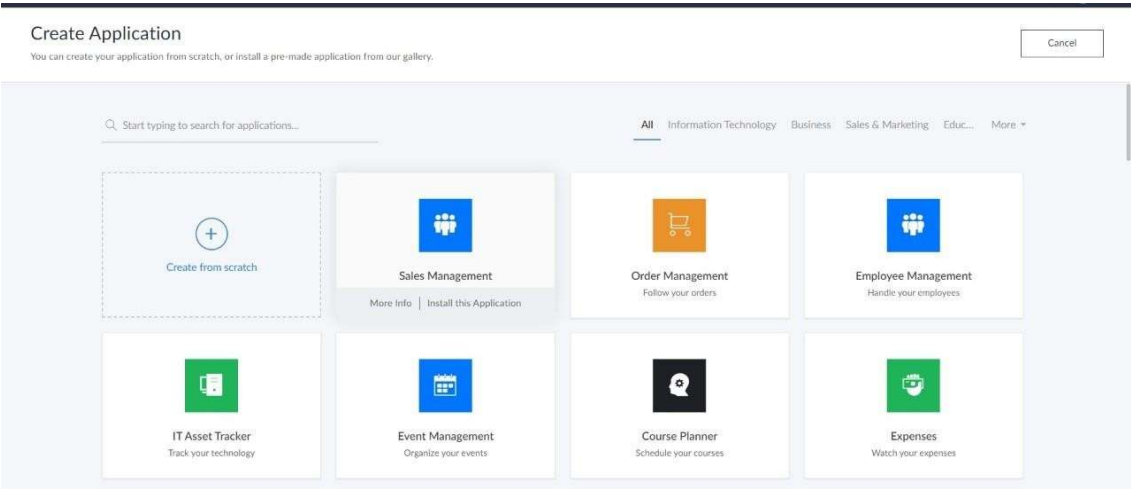
STEP1: GOTO ZOHOCOM



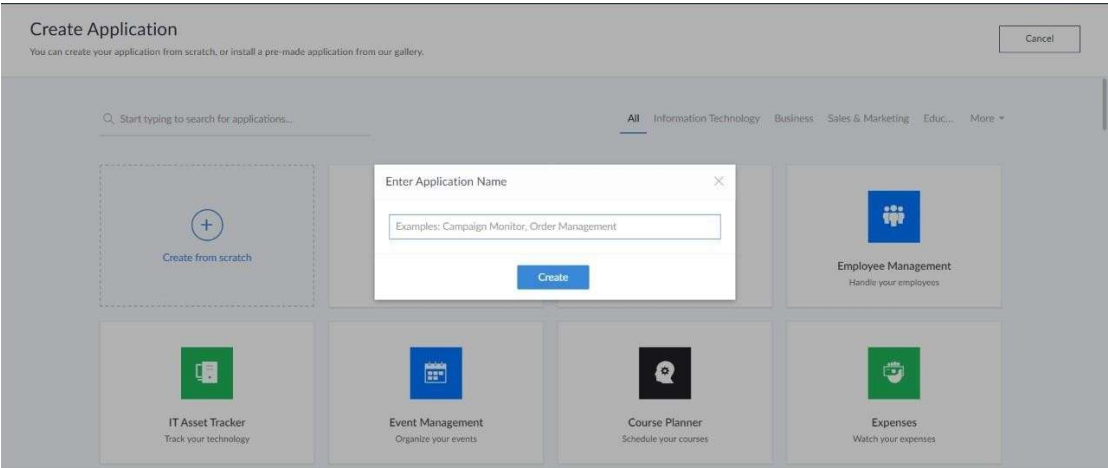
STEP 2: LOGINTOTHE ZOHOCOM



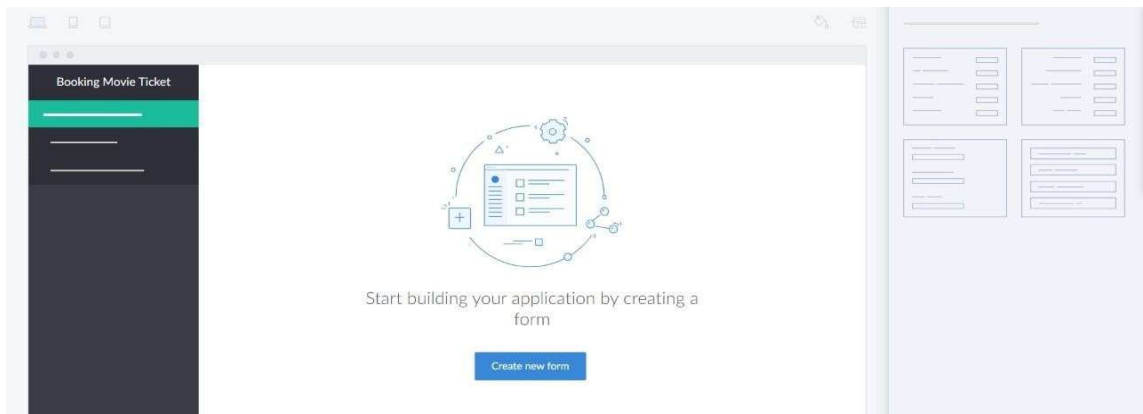
STEP 3: SELECT ONE APPLICATION



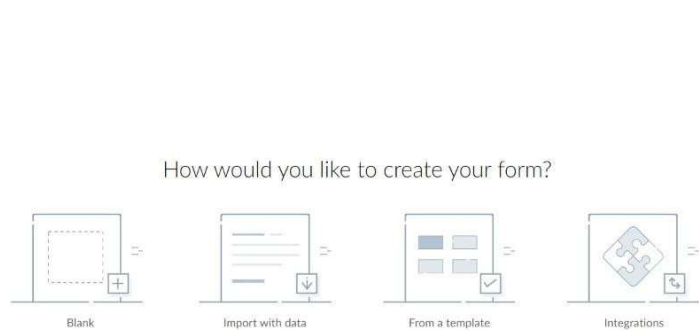
STEP 4: ENTER APPLICATION NAME



STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.

PSA

PROPERTY BUYING AND RETAIL

rental

rental

rental Report

Kanishk

rental

Name

First Name

Last Name

Email

Address

Address Line 1

Address Line 2

City / District

State / Province

Postal Code

Country

Currency

###,###,###

Drop Down

-Select-

Submit

Booking Movie Ticket
user details

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

Name

Phone

Email

Date-Time

Theatres

Drop Down

Field name

Name

Field link name

Name

Validation

Mandatory

Display Fields

Prefix

First Name

Last Name

Suffix

Data Privacy

PROPERTY BUYING
AND RETAIL

rental

rental Report

rental Report

Name	Email	Address	Currency	Drop Down
KANISHK C	kanicoco1756@gmail.com	1/98 hello street, saveetha nagar, chennai, TN, 600053, India	₹ 56,000.00	RENT PAID

EXP NO 4: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR CAR BOOKING RESERVATION SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

DATE:

AIM:

To Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as Car Booking Reservation System.

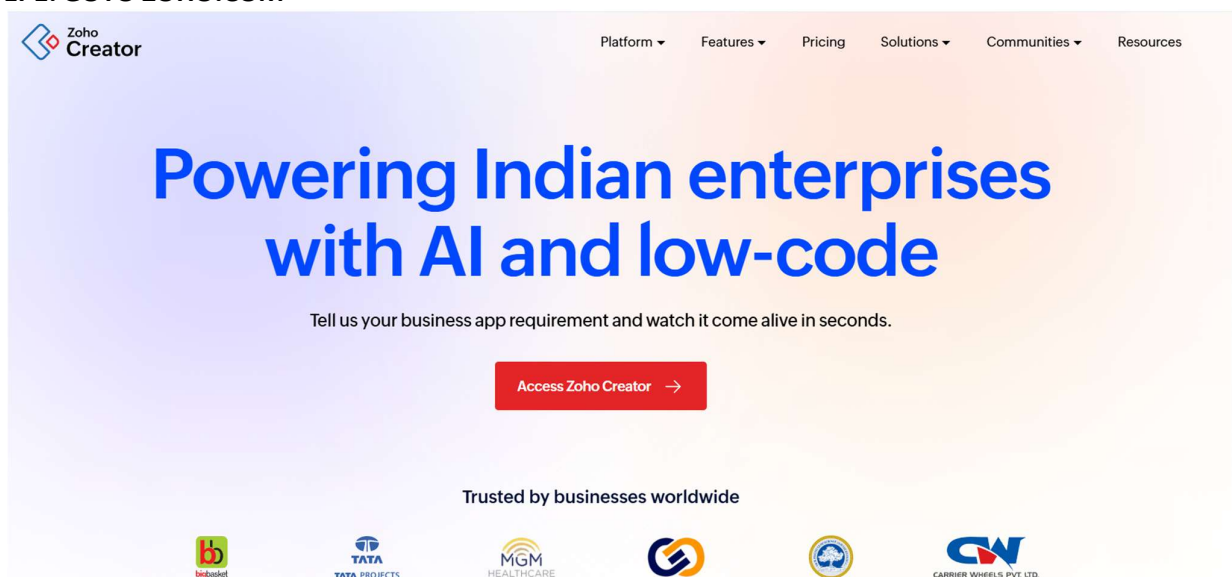
step 5: Created new application as Car Booking Reservation System.

step 6: Select one form

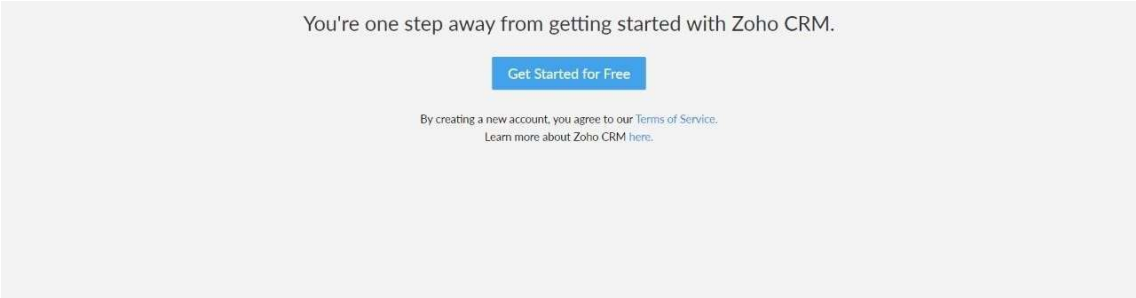
step 7: The software has been created.

IMPLEMENTATION:

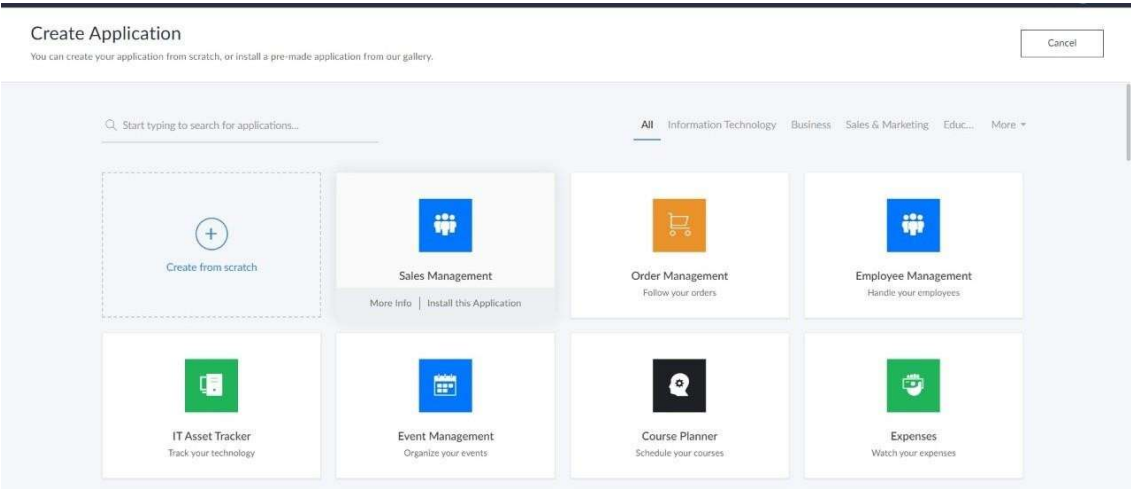
STEP1: GOTO ZOHO.COM



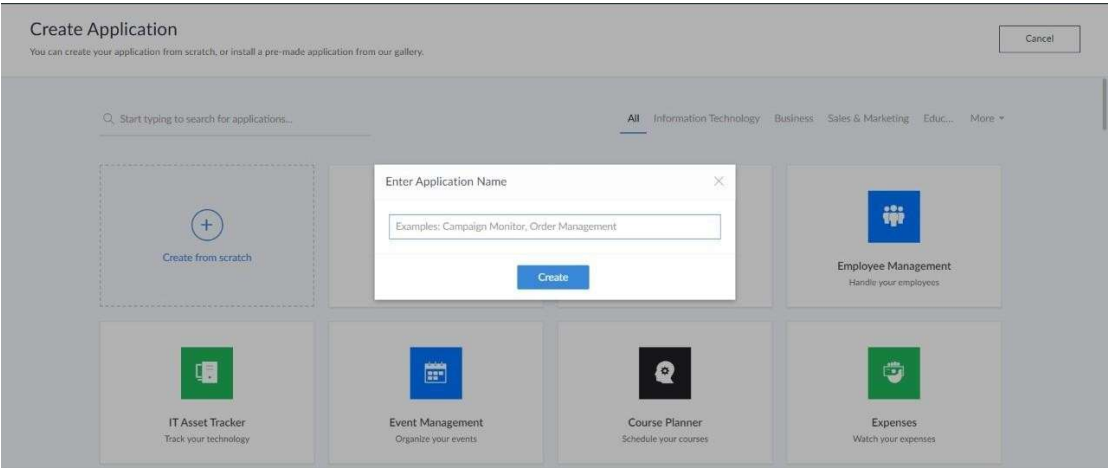
STEP 2: LOGINTOTHE ZOHO.COM



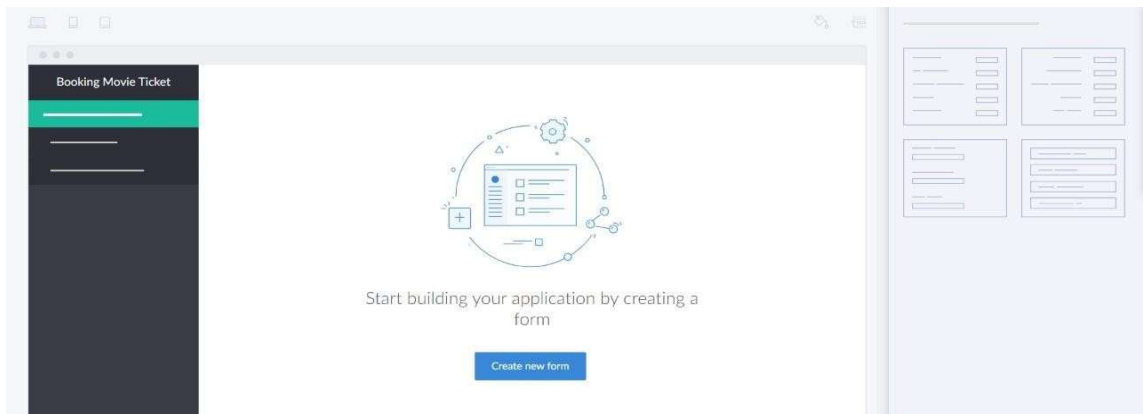
STEP 3: SELECT ONE APPLICATION



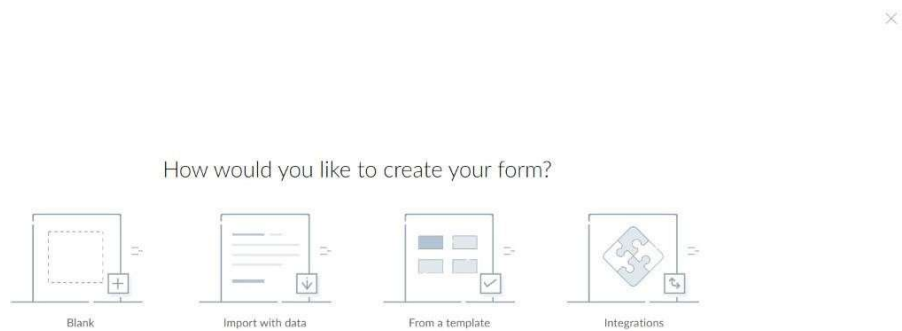
STEP 4: ENTER APPLICATION NAME



STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.

CAR RESERVATION
Book a Cab

Basic Fields

Name	Email
Address	Phone
Single Line	Multi Line
123 Number	15 Date
Time	Drop Down

Name *

Employee ID *

Email Id *

Mobile Number *

Gender *

Booking Date *

Travel Date *

Pickup Time *

Pickup Address *

Destination Address *

 CAR RESERVATION

Development

CR

Book a Cab

Book a Cab

Name *

First NameLast Name

Employee ID *

Email Id *

Mobile Number *

 +91

81234 56789

Gender *

☐ Male

☐ Female

Booking Date *

31-Jul-2026

Travel Date *

dd-MMM-yyyy

Pickup Time *

-Select-

Pickup Address *

EXP NO 5: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR LIBRARY BOOK RESERVATION SYSTEM FOR SIMATS LIBRARY USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS

DATE:

AIM:

To Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as library book reservation system.

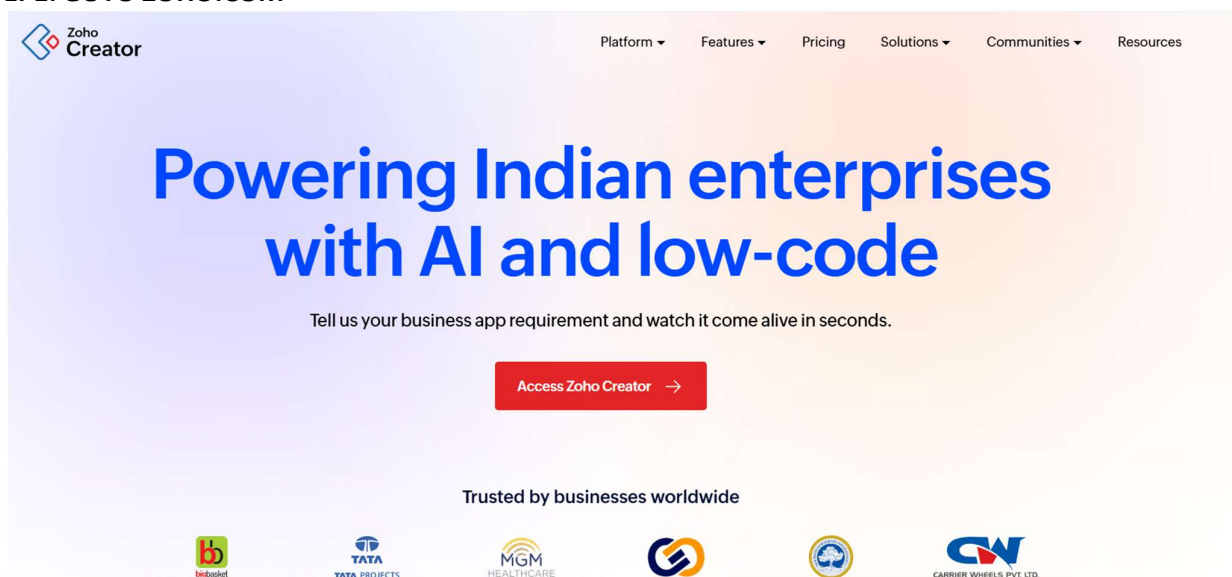
step 5: Created new application as library book reservation system.

step 6: Select one form

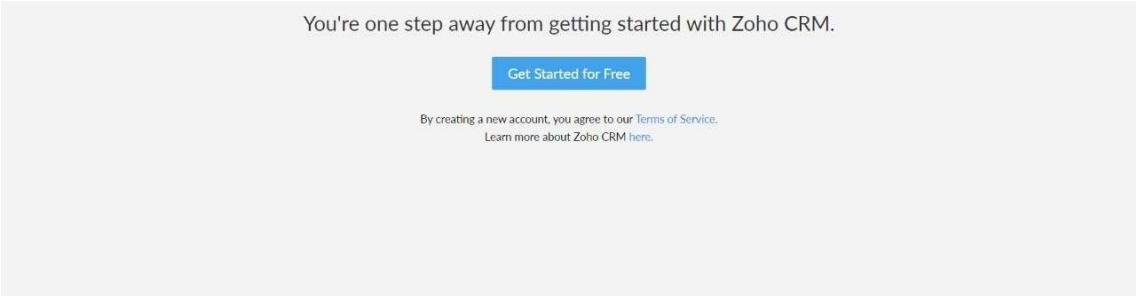
step 7: The software has been created.

IMPLEMENTATION:

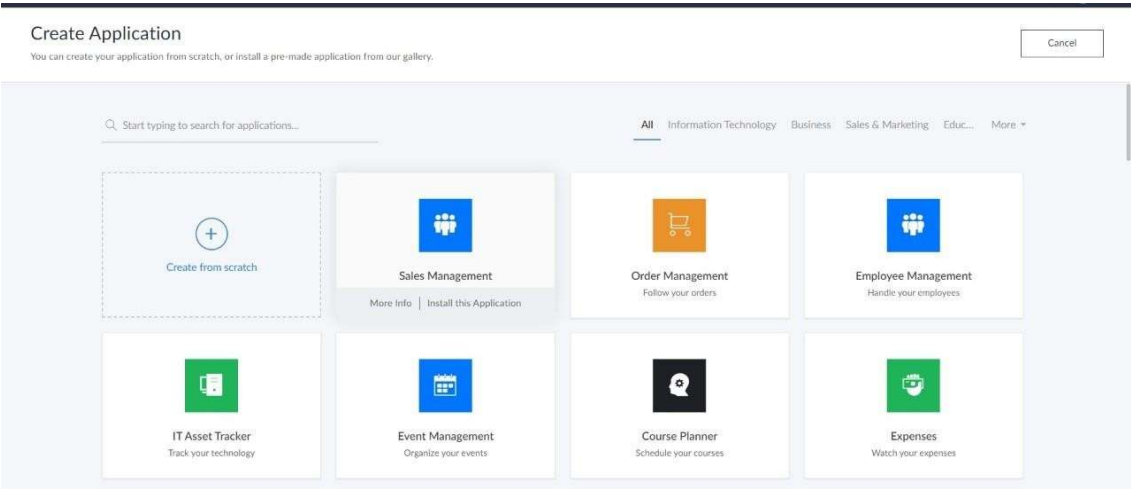
STEP1: GOTO ZOHOCOM



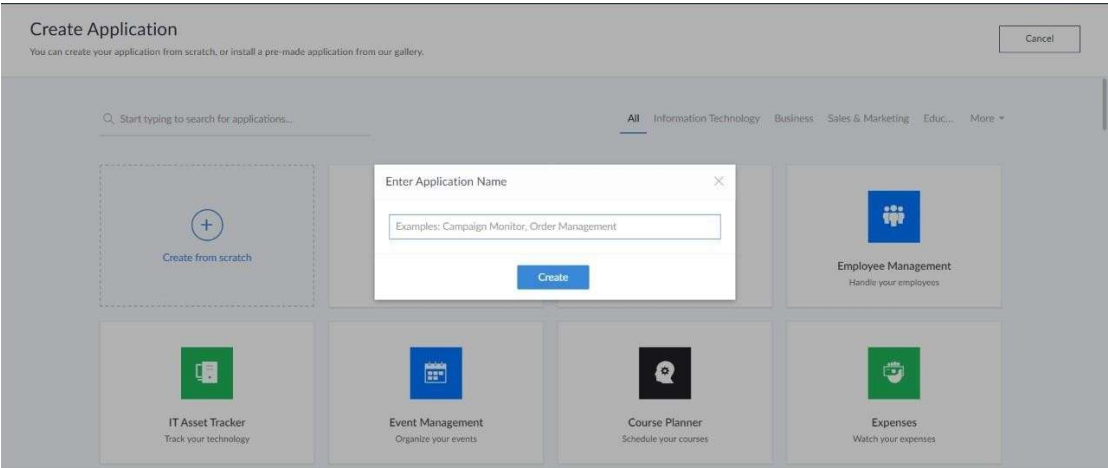
STEP 2: LOGIN TO THE ZOHO.COM



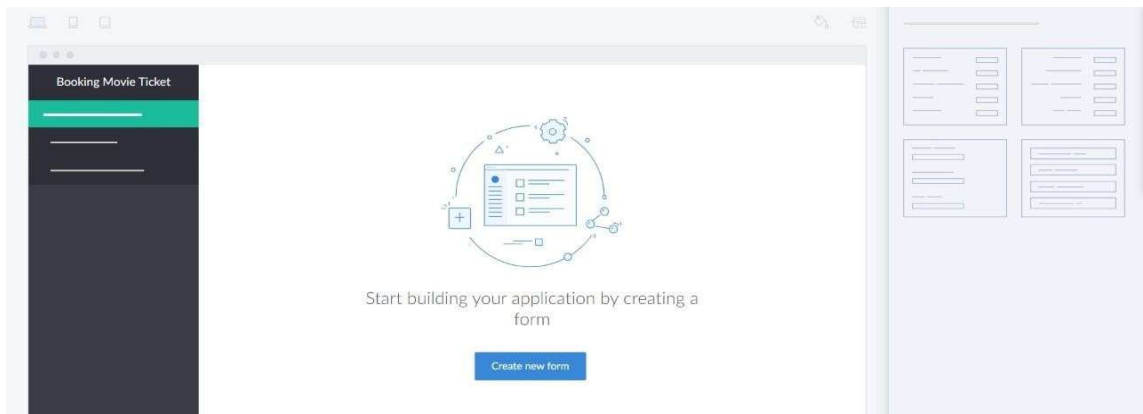
STEP 3: SELECT ONE APPLICATION



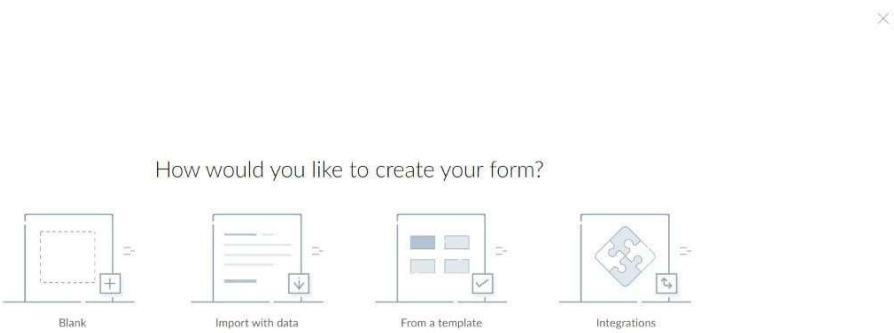
STEP 4: ENTER APPLICATION NAME



STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.

LIBRARY MANAGEMENT

Membership Form

+

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Time

Drop Down

Name *

ID *

Date of birth

Your notes content will be shown here

Occupation *

Phone Number

Email *

☒

What would you use the library for? *

☒

Which sections of the library would you like access to?

☐

Access to E-Resources

LIBRARY MANAGEMENT

Development

View as

-MySe

LM

LIBRARY MANAGEMENT

Membership Form

Membership Form

Registered Members

Membership Form

Name *

KANISHK

C

First NameLast Name

ID *

ABC123

Date of birth

27-May-2024

Occupation *

Student

Phone Number

+91 8778470272

Email *

kanicoco1756@gmail.com

What would you use the library for? *

☒ Reference

☐ In-house reading

☐ Borrowing

Which sections of the library would you like access to?

☒ All

☐ Magazines

☐ Fiction

☐ Non-Fiction

☐ Electronic

☐ Research & Reference

LIBRARY MANAGEMENT

Membership Form

Registered Members

ID	Name	Occupation	Email	Phone Number
ABC123	KANISHK C	Student	kanicoco1756@gmail.com	+918778470272

EXP NO 6: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR PRODUCT SELLING USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

DATE:

AIM:

To create a simple cloud software application for product selling using any cloud service provider to demonstrate saas.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as product selling.

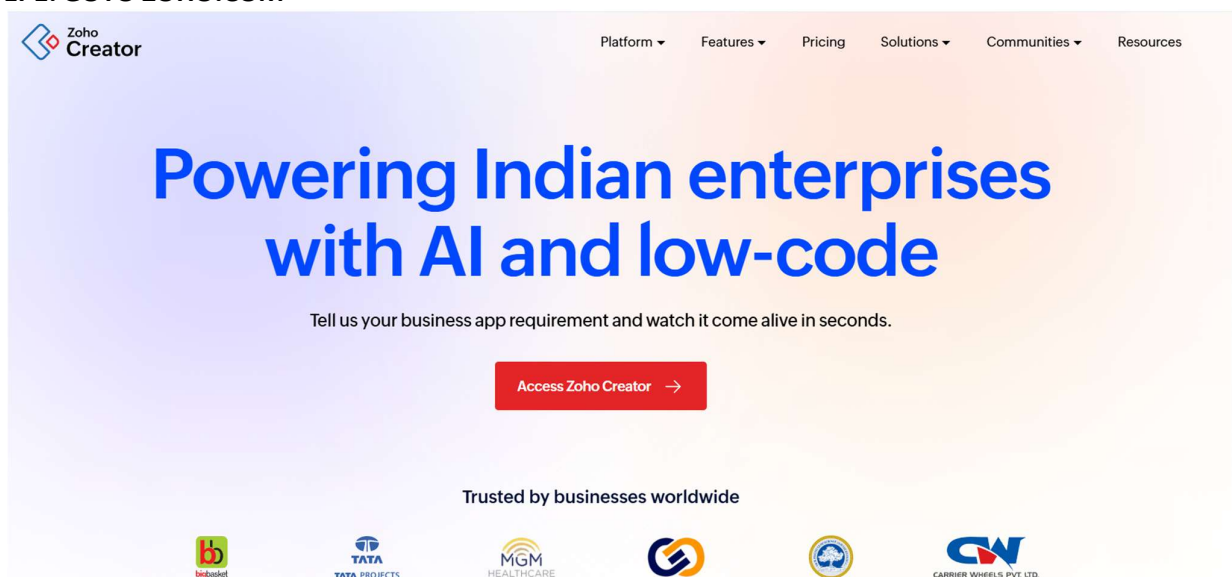
step 5: Created new application as product selling.

step 6: Select one form

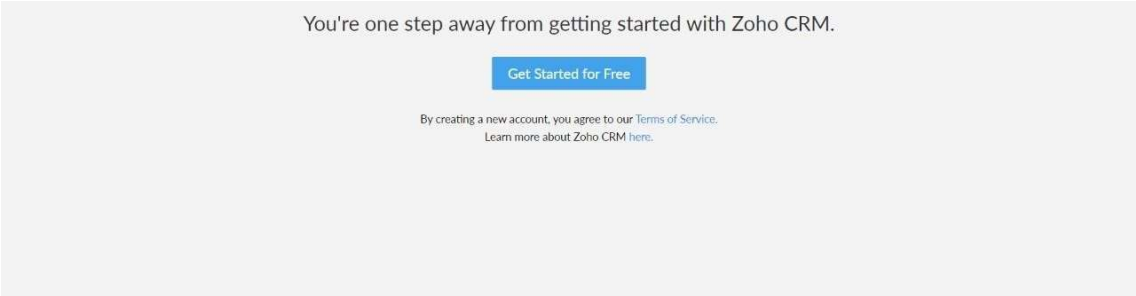
step 7: The software has been created.

IMPLEMENTATION:

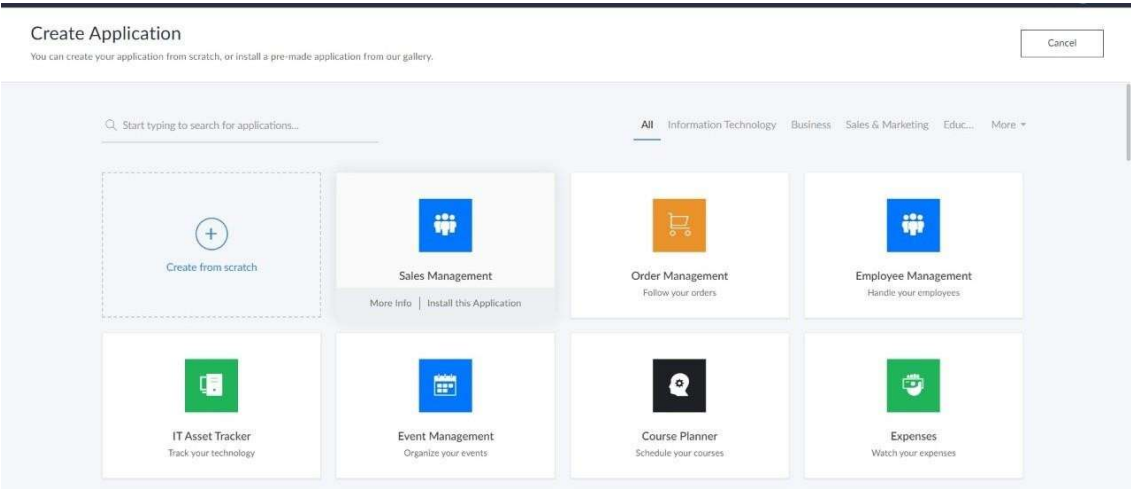
STEP1: GOTO ZOHO.COM



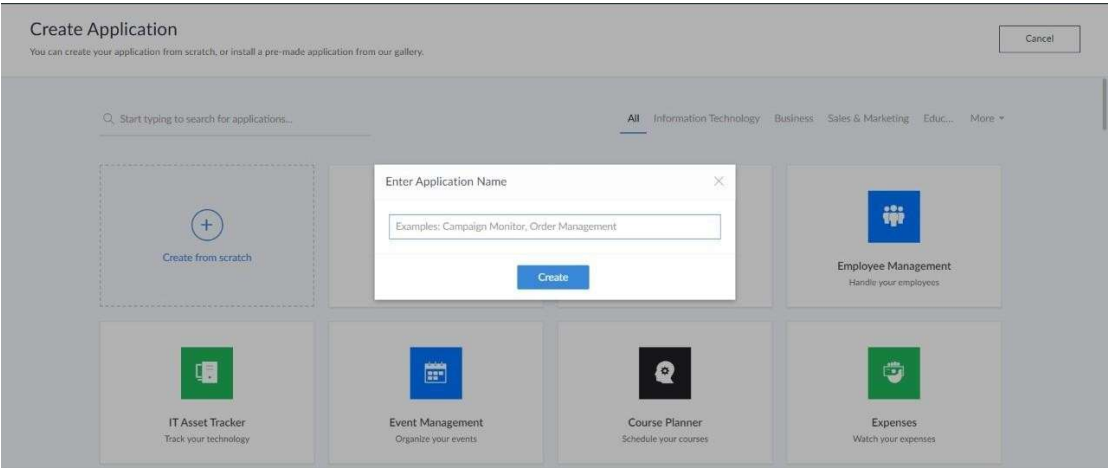
STEP 2: LOGIN TO THE ZOHO.COM



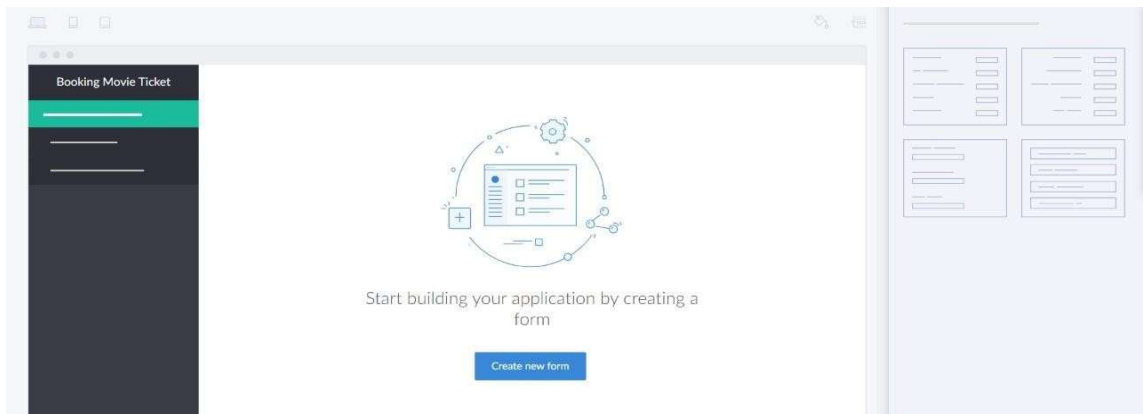
STEP 3: SELECT ONE APPLICATION



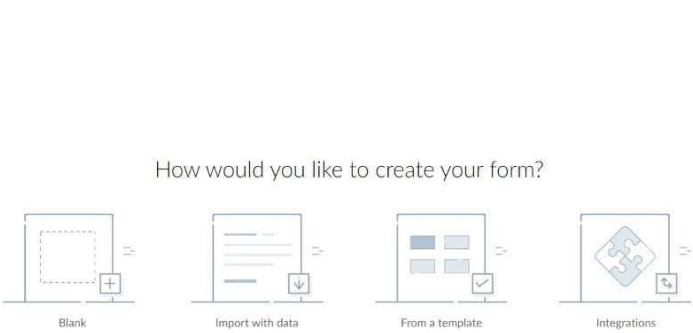
STEP 4: ENTER APPLICATION NAME



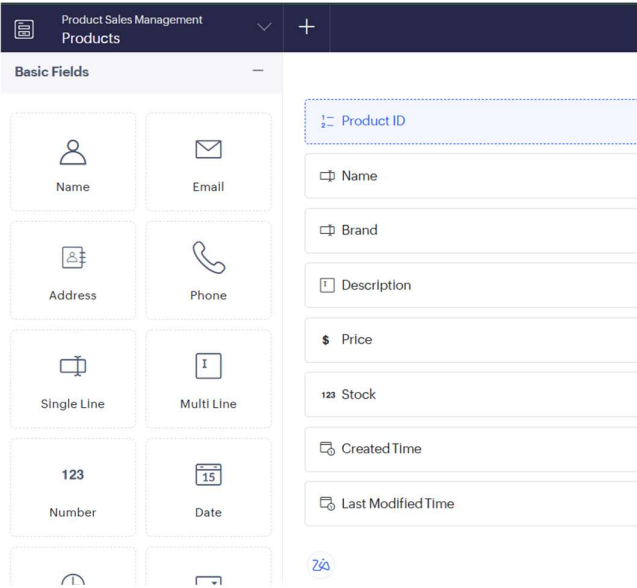
STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.



Product Sales Management

Product Sales Management

Products Report

		Product ID	Name	Brand	Description	Price	Stock	Created Time	Loc
		5	Webcam HD	VisionTech	Crystal clear video quality	\$ 59.99	90	19-Jul-2026 11:00:00	19
		4	Monitor Stand	DeskPro	Adjustable height support system	\$ 34.99	120	18-Jul-2026 13:20:00	18
		3	Mechanical Keyboard	KeyMaster	RGB backlit typing experience	\$ 89.99	200	17-Jul-2026 08:00:00	17
		2	USB-C Hub	TechBrand	Multi-port connectivity solution	\$ 45.50	75	16-Jul-2026 10:15:00	16
		1	Wireless Mouse	TechBrand	Ergonomic design for comfort	\$ 29.99	150	15-Jul-2026 09:30:00	15

Trial has expired

Upgrade

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Dashboard

Inventory Updates

Availability Checks

Sales Transactions

Products

Products

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